

Exp No:16 SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

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PROGRAM:

The screenshot displays the Keil uVision IDE interface. The top menu bar includes File, Edit, View, Project, Flash, Debug, Peripherals, Tools, SVCS, Window, and Help. Below the menu is a toolbar with various icons for file operations and debugging. The left pane shows the Project Explorer with 'Project: EXP16' and 'Target 1'. The main editor window displays the source code for 'exp16.c'. The code includes headers for `reg51.h` and `stdio.h`, defines a serial interrupt service routine (`serial_ISR`), and a delay function (`delay`). The `main` function initializes the serial port (SCON, TMOD, TH1, TR1, IE, TI) and enters a loop that prints 'Hello World' and calls the delay function. The bottom pane shows the Build Output window with the following text:

```
ADDRESS: 080AH
Program Size: data=9.0 xdata=0 code=15
".\Objects\EXP16" - 0 Error(s), 2 Warning(s).
Build Time Elapsed: 00:00:01
```

A) DISPLAYING A MESSAGE

The screenshot shows the Keil uVision IDE with the following components:

- Registers:** A table showing the state of registers. The PC register is at address 0x0CA7. The SP register is at 0x4b.
- Disassembly:** The assembly code for the program. The instruction at address 0x0CA7 is highlighted: `MOV SC0N(0x98), #0x50`.
- exp16.c:** The C source code. The `main` function is shown, which initializes `SCON=0x50`, sets `TMOD=0x20`, `TH1=0xFD`, and enters a `while(1)` loop. Inside the loop, it calls `printf("Hello World\n");` and `delay();`.
- Command:** The command window shows the execution command: `Load "C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\Objects\EXP16"`.
- UART #1:** The serial console window displays the output: `Hello World` repeated five times.

B) DURING INTERRUPTION:

The screenshot shows the Keil uVision IDE with the following components:

- Registers:** A table showing the state of registers. The PC register is at address 0x0CA7. The SP register is at 0x4b.
- Disassembly:** The assembly code for the program. The instruction at address 0x0CA7 is highlighted: `MOV SC0N(0x98), #0x50`.
- exp16.c:** The C source code. The `main` function is shown, which initializes `SCON=0x50`, sets `TMOD=0x20`, `TH1=0xFD`, and enters a `while(1)` loop. Inside the loop, it calls `printf("Hello World\n");` and `delay();`.
- Command:** The command window shows the execution command: `Load "C:\Keil_v5\C51\Examples\HELLO\EMBEDDED C\EXP11\Objects\EXP16"`.
- UART #1:** The serial console window displays the output: `>>> INTERRUPT RECEIVED: h`, `>>> INTERRUPT RECEIVED: d`, `>>> INTERRUPT RECEIVED: y`, and `>>> INTERRUPT RECEIVED: u`.
- Simulation:** The bottom status bar indicates the simulation is running.